Precision Enteral Feeding & Somatic Growth for Preterm Infants

A deterministic bedside decision-support platform designed for neonatologists, pediatricians, and clinical feeding teams. Replaces complex mental math with instant, zero-error prescription volume, scoop dilution, and ESPGHAN 2022 accordance.

THE CLINICAL CHALLENGE IN THE NICU VS. THE NEOPED™ SOLUTION

⚠ The Daily Bedside Pain Points

- **Time-Consuming Manual Calculations:** Converting weight (g) and fluid allowance (mL/kg/d) into daily volume, powder grams, scoops, water dilution, and hourly rates takes 10–15 minutes per preterm baby.
- **Risk of Bedside Arithmetic Error:** Dilution mistakes can cause hyperosmolar stress, necrotizing enterocolitis (NEC) risk, or inadequate protein accretion.
- **Guideline Disconnect:** Manually auditing energy, 60:40 protein, lipids, and calcium/phosphorus ratios against ESPGHAN 2022 recommendations during rounds is rarely practical.

✓ How NeoPed™ Solves It for Doctors

- **1-Step Input in Under 10 Seconds:** Simply enter Patient Weight and Fluid Goal — instantly receive total volume, exact scoops of Pediamil® LBW, and sterile water.
- **Automated Feeding Intervals:** Ready-to-administer volumes and scoops for 8 feeds (q3h), 12 feeds (q2h), or syringe pump continuous infusion.
- **Live ESPGHAN 2022 Compliance:** Every nutrient evaluated dynamically against peer-reviewed preterm standards.
- **1-Click Hospital Feeding Sheet (PDF):** Complete, printable order form with attending physician signature block ready for patient records.

THE 4-STEP CLINICAL JOURNEY (HOW DOCTORS USE IT)

1 Patient Parameters Input

The clinician enters the infant's **Weight in grams** (e.g., 1,350g VLBW) and **Target Fluid Allowance** (135–200 mL/kg/day, typical 150–160 mL/kg/d). Quick preset buttons allow rapid selection during ward rounds.

Clinician Prescription Supremacy: The doctor's prescribed volume always governs feeding orders.

2 Instant Scoops & Volume Engine

The engine applies standard Pediamil® LBW dilution (15.0% w/v = 3 level scoops [15g] + 90 mL water = 100 mL feed) to output:

TOTAL 24H VOLUME (ML)	EXACT SCOOPS/DAY	STERILE WATER (ML)

3 Live ESPGHAN 2022 Accretion Audit

Evaluates delivered nutrients in real time: Energy (115–140 kcal/kg/d), True Protein (3.5–4.0 g/kg/d), P:E Ratio (2.8–3.6 g/100 kcal), Lipids (4.8–8.1 g/kg/d), Ca, P, Fe, and Vitamin D3 with instant visual coverage badges.

✓ All core nutrients covered at standard 150–160 mL/kg/day feeding volumes.

4 Bedside Schedule, Growth & PDF Sheet

Generates bedside administration timetables (volume & scoops per feed), plots weight/length/head circumference on **Fenton 2013 Preterm Curves** (with LMS Z-scores), and exports a professional **Hospital Feed Sheet PDF**.

WHY KEY OPINION LEADERS RECOMMEND NEOPED™

100% Deterministic Math

Zero guessing, rounding drift, or human memory errors. Verified by 227 automated acceptance tests.

Mobile-First Ward Tool

Doctors can calculate feeds directly on their phone at the bedside or on workstation monitors.

100% Client-Side Privacy

Zero Protected Health Information (PHI) leaves the device. Complete HIPAA and GDPR compliance.

WHAT DOES THE PRESCRIBED VOLUME & SCOOPS DELIVER TO THE INFANT?

When an attending physician prescribes **Pediamil® LBW** at standard physiological fluid volumes (150–160 mL/kg/day), the infant receives the exact enteral accretion required for intrauterine-like catch-up growth without fluid or solute overload.

Delivered Energy

119.6 kcal/kg/d

at 150 mL/kg/day

ESPGHAN Target: 115–140 kcal/kg/d. Fueling high preterm metabolic rate.

True Protein

3.63 g/kg/d

60:40 Whey:Casein

ESPGHAN Target: 3.5–4.0 g/kg/d. High α-lactalbumin promotes lean muscle accretion.

P:E Ratio

3.04 g / 100 kcal

Optimal Anabolic Range

ESPGHAN Target: 2.8–3.6 g/100 kcal. Promotes nitrogen balance without acidosis.

Delivered Lipids

7.32 g/kg/d

>60% Milk Fat + LCPUFA

ESPGHAN Target: 4.8–8.1 g/kg/d. Easily digested milk fat + DHA/ARA (1:1).

PEDIAMIL® LBW SPECIFICATION VS. ESPGHAN 2022 RECOMMENDATIONS

NUTRIENT / PARAMETER	PEDIAMIL® LBW (/100G POWDER)	PEDIAMIL® LBW (/100 ML FEED)	DELIVERED DOSE (150 ML/KG/D)	DELIVERED DOSE (160 ML/KG/D)	ESPGHAN 2022 PRETERM TARGET	STATUS
Energy	531 kcal	79.7 kcal	119.6 kcal/kg/d	127.5 kcal/kg/d	115–140 kcal/kg/d (Cond. 140–160)	TARGET MET ✓
True Protein (60:40)	16.1 g (9.66g Whey)	2.42 g (1.45g Whey)	3.63 g/kg/d	3.87 g/kg/d	3.5–4.0 g/kg/d (Cond. up to 4.5)	TARGET MET ✓
Protein-to-Energy (P:E)	3.04 g/100 kcal	3.04 g/100 kcal	3.04 g/100 kcal	3.04 g/100 kcal	2.8–3.6 g / 100 kcal	OPTIMAL RANGE ✓
Total Lipids (Fat)	32.5 g (Milk Fat)	4.88 g	7.32 g/kg/d	7.81 g/kg/d	4.8–8.1 g/kg/d	TARGET MET ✓
DHA & ARA (1:1 Ratio)	90 mg / 90 mg	13.5 mg / 13.5 mg	20.3 mg/kg/d each	21.6 mg/kg/d each	12–30 mg/100 kcal (1:1 Balanced)	BALANCED 1:1 ✓
Calcium (Ca^{2+})	620 mg	93.0 mg (2.32 mmol)	139.5 mg/kg/d	148.8 mg/kg/d	120–140 mg/kg/d (3.0–3.5 mmol/kg/d)	TARGET MET ✓
Phosphorus (P)	355 mg	53.3 mg (1.72 mmol)	80.0 mg/kg/d	85.3 mg/kg/d	60–90 mg/kg/d (Ca:P 1.6–2.0:1)	CA:P 1.75:1 ✓
Elemental Iron (Fe)	10.5 mg	1.58 mg	2.37 mg/kg/d	2.53 mg/kg/d	2.0–3.0 mg/kg/d (Prophylactic Fe)	ENTERAL FE MET ✓
Vitamin D3	11.1 µg (444 IU)	1.66 µg (66.4 IU)	99.6 IU/kg/d	106.2 IU/kg/d	400–1,000 IU/day absolute intake	ADD DROPS IF <400 IU
Sodium (Na^{+})	320 mg	48.0 mg (2.09 mmol)	3.13 mmol/kg/d	3.34 mmol/kg/d	3.0–5.0 mmol/kg/d (ELBW up to 8.0)	ELECTROLYTE MET ✓
Potassium (K^{+})	575 mg	86.3 mg (2.21 mmol)	3.31 mmol/kg/d	3.53 mmol/kg/d	2.0–3.5 mmol/kg/d (78–137 mg/kg/d)	ELECTROLYTE MET ✓
Chloride (Cl^{-})	500 mg	75.0 mg (2.12 mmol)	3.17 mmol/kg/d	3.39 mmol/kg/d	3.0–5.0 mmol/kg/d (106–177 mg/kg/d)	ELECTROLYTE MET ✓
Magnesium (Mg^{2+})	50 mg	7.5 mg	11.3 mg/kg/d	12.0 mg/kg/d	8.0–15.0 mg/kg/d	TARGET MET ✓
Zinc (Zn^{2+})	6.0 mg	0.90 mg	1.35 mg/kg/d	1.44 mg/kg/d	1.4–2.5 mg/kg/d	TARGET MET ✓

Key Opinion Leader Takeaway: Standalone Enteral Adequacy

Pediamil® LBW provides a complete, balanced nutritional profile that directly satisfies ESPGHAN 2022 recommendations across macronutrients, electrolytes, and bone minerals at standard enteral fluid volumes. **Doctors do not need to add routine protein fortifiers or electrolyte modulars** unless specific clinical restrictions apply, minimizing handling and contamination risks in the NICU.

STANDARD PREPARATION & RECONSTITUTION PROTOCOL

1. Standard Dilution

15.0% w/v concentration

15.0g powder in 90 mL sterile water yields **100 mL prepared feed** (caloric density 79.7 kcal/100 mL).

2. Scoop Calibration

5.0 grams per level scoop

Use only the enclosed manufacturer scoop. Level with a sterile spatula without compressing or packing powder.

3. Safe Reconstitution

> 70°C sterile boiled water

Dissolve powder in sterile water >70°C per WHO/FAO PIF guidelines. Cool to feeding temperature before administration.

BEDSIDE FEEDING TIMETABLE: PRACTICAL CLINICAL CASE EXAMPLE

Patient Case: VLBW Preterm Infant, Weight: **1,350g (1.350 kg)** • Fluid Allowance: **150 mL/kg/day** • Total 24h Volume: **203 mL/day** • Daily Powder: **30.5g** (~**6.1 scoops/day** in **182 mL sterile water**).

ADMINISTRATION ROUTE / REGIMEN	FREQUENCY / 24H	VOLUME PER FEED	PEDIAMIL® LBW SCOOPS	NURSING BEDSIDE ADMINISTRATION NOTE
q3h Enteral Bolus Feeding Standard NICU convalescent schedule	8 feeds / day	25.4 mL / feed	~0.8 scoop / feed	Administer by gravity bolus over 15–20 minutes. Check gastric residual volume prior to feed if clinically indicated.
q2h Enteral Bolus Feeding For infants with smaller gastric capacity	12 feeds / day	16.9 mL / feed	~0.5 scoop / feed	Smaller, frequent aliquots reduce gastric distension and prevent reflux episodes in high-risk preterm infants.
Continuous Syringe Pump Infusion Severe feeding intolerance or micro-preemie	Continuous 24h	8.5 mL / hour	–	Infuse via enteral syringe pump over 24 hours. Invert syringe periodically to prevent lipid phase separation.

DUAL GROWTH TRAJECTORY ENGINE: FENTON 2013 & WHO 2006 STANDARDS

1. Fenton 2013 Preterm Growth Curves (22–50w PMA)

• **Cole & Green LMS Z-Score Interpolation:** Calculates exact standard deviation scores (Z) and centiles (10th, 50th, 90th) for **Weight, Length, and Head Circumference** from 22 completed gestational weeks up to 50 weeks Post-Menstrual Age (PMA).
• **Extreme Range Precision:** Accurately formats extreme bounds (<0.1th percentile and >99.9th percentile) without mathematical distortion.
• **Identifies Extrauterine Growth Restriction (EUGR):** Alerts clinicians if weight drops below the 10th percentile.

2. Seamless WHO 2006 Transition & Graduation

• **Automatic WHO 2006 Transition:** Once the infant reaches term equivalent age (>50 weeks PMA), growth plotting automatically transitions to **WHO Child Growth Standards (0–24 months Corrected Age)**.
• **Pediamil® 1 Term Step-Down Milestone:** When the infant crosses **3,500g** (term-equivalent body mass), the engine automatically suggests stepping down from LBW formula to **Pediamil® 1** (68.5 kcal, 1.49g protein/100 mL, 13.7% dilution) to prevent excessive protein and renal solute burden.

Verified Against Chou et al. (PediTools) Benchmarks:
100% concordance across all 37 official growth dataset checkpoints (Fenton 2013 & WHO 2006 LMS tables).

100% VALIDATED LMS MATH

THE ONE-CLICK HOSPITAL FEEDING ORDER SHEET (PDF DOWNLOAD)

Senior pediatricians and neonatologists requested an exportable, official medical document that can be attached directly to the patient's paper chart, entered into hospital electronic health records (EHR), or handed to the nursing team during shift changes.

OFFICIAL 7-SECTION HOSPITAL CLINICAL ORDER SHEET CONTAINS:

READY FOR PHYSICIAN SIGNATURE

- **Section 1 — Institutional Header:** Hospital name, Liptis Nutrition logo, and medical protocol title.
- **Section 2 — Patient Demographics:** Weight (g/kg), Gestational Age, Chronological Age, and Corrected Age.
- **Section 3 — Physician Order:** Prescribed fluid allowance (mL/kg/d) and calculated 24h total feed volume.
- **Section 4 — Preparation Recipe:** Total powder mass (g), exact scoops, sterile water volume, and dilution.
- **Section 5 — Bedside Feeding Timetable:** Exact mL per feed for q3h (8 feeds), q2h (12 feeds), or continuous pump.
- **Section 6 — ESPGHAN 2022 Accretion Table:** Delivered energy, protein, lipids, Ca, P, Fe, and vitamins.
- **Section 7 — Medical Attestation Block:** Attending physician signature line, stamp box, date, time, and medical license number.

AUTHORITATIVE PEER-REVIEWED EVIDENCE REGISTRY

CLINICAL GUIDELINE / STANDARD	PRIMARY CITATION & PUBLICATION	KEY EVIDENCE ROLE IN NEOPED™	VERIFIED INDEX
ESPGHAN 2022 Preterm Enteral Position Paper	Embleton ND, Moltu SJ, Lapillonne A, et al. *J Pediatr Gastroenterol Nutr*. 2023;76(2):248–268.	Defines all enteral energy (115–140 kcal/kg/d), protein (3.5–4.5 g/kg/d), lipid, mineral, and electrolyte bounds.	PubMed PMID: 36705703
Fenton 2013 Preterm Growth Curves	Fenton TR, Kim JH. *BMC Pediatrics*. 2013;13:59.	Intrauterine preterm growth reference curves from 22 to 50 weeks PMA with exact LMS parameters.	PubMed PMID: 23601190
WHO 2006 Child Growth Standards	WHO Multicentre Growth Reference Study Group. *Acta Paediatr Suppl*. 2006;450:76–85.	International gold-standard growth trajectories for term-equivalent infants from 0 to 24 months Corrected Age.	PubMed PMID: 16817681
Cole & Green (1992) LMS Method	Cole TJ, Green PJ. *Stat Med*. 1992;11(10):1305–1319.	Mathematical Box-Cox power transformation for precise age-adjusted Z-score and percentile derivation.	PubMed PMID: 1518992
PediTools Independent Benchmarks	Chou JH, Roumiantsev S, Singh R. *J Med Internet Res*. 2020;22(1):e16204.	Independent clinical verification benchmark used to validate NeoPed™ LMS percentile calculations.	PubMed PMID: 32012066
WHO/FAO PIF Guidelines	World Health Organization & FAO (Geneva, 2007) ISBN 978-92-4-159541-4.	Safe reconstitution standards (>70°C sterile water) for powdered infant formula in healthcare facilities.	WHO Item 9789241595414
IUPAC Standard Atomic Weights	Prohaska T, et al. (CIAAW) *Pure Appl Chem*. 2022;94(5):573–600.	Standard atomic weights for stoichiometric molar conversions (Na 22.99, K 39.10, Cl 35.45, Ca 40.08, P 30.97).	DOI: 10.1515/pac-2019-0603

Institutional Medical Governance & Clinical Decision Support Notice:

Liptis Nutrition NeoPed™ is an institutional Clinical Decision Support (CDS) reference calculator intended solely to assist licensed neonatologists, pediatricians, and clinical nutritionists in evaluating enteral feeding regimens. It does not replace individualized clinical judgment, bedside physiological monitoring, or hospital laboratory evaluations. All calculations are executed 100% locally within the user's browser runtime with zero network egress of patient identifiable data.

PREPARED FOR KEY OPINION LEADERS & NICU FACULTY BY:

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CLINICAL VERIFICATION ATTESTATION:

227 / 227 Acceptance Tests Passed (100.0%)
Fully Concordant with ESPGHAN 2022 & Fenton 2013